conviction_tilt

equity strategy | family: equity-neutralized | status: validated | derived from `graded_bets`

What this strategy does

Skip the middle 15% either side of average, lean toward the strongest surprises, balance by size

Why it should work. Concentrate capital where the surprise is largest; neutralize size so it is not a hidden bet.

Design notes (from the strategy's source file).

Conviction Tilt -- skip the unsure middle, lean on the strongest signals, balance by size.

Built after diagnosing why rank-weighting had the better Sharpe but the worse return: it traded ~4x more positions and saturated the liquidity/leverage caps. This strategy drops the low-conviction middle of the SUE-rank distribution (trim 0.15), leans weight toward the extremes (tilt), keeps size neutrality, and when the leverage cap binds drops the LOWEST-conviction positions first instead of shaving everything equally.

parameter	value
base_unit_fraction	0.002
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

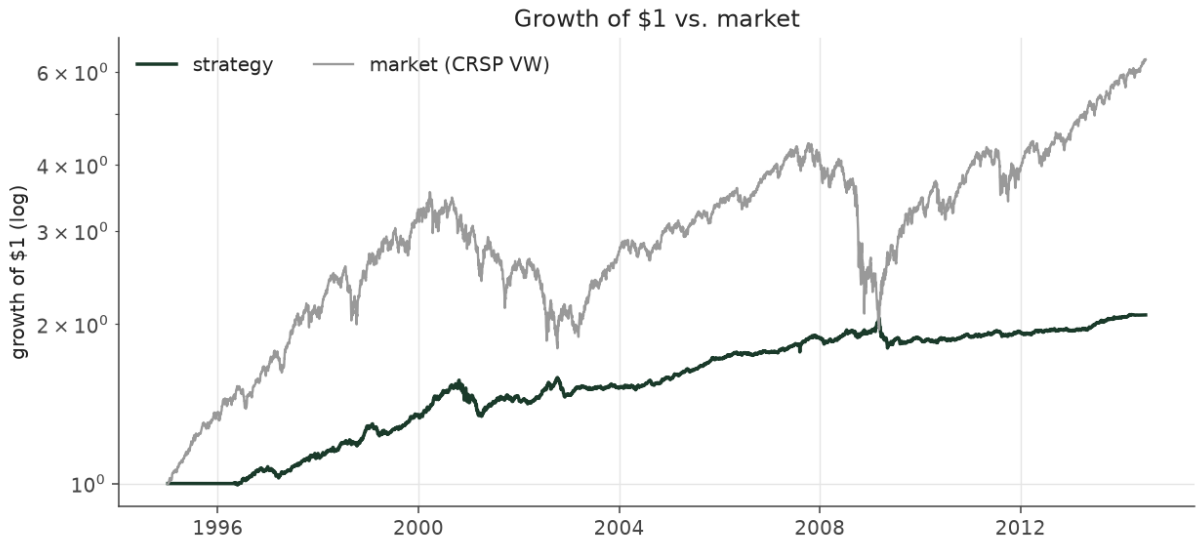
metric	conviction tilt	market (CRSP VW)
Annualized return	3.84%	9.94%
Annualized volatility	3.97%	19.35%
Sharpe (NAV-pct)	0.97	0.59
Sortino	1.38	
Max drawdown	-14.5%	-55.5%
Calmar	0.26	
Final NAV ($10M start)	$20,828,564	
Worst day	-2.23%	
% positive days	50.8%	
Sharpe 1996-2001	1.24	
Sharpe 2002-2007	1.35	
Sharpe 2008-2013	0.41	

Correlation of daily returns with the market: **0.05**.

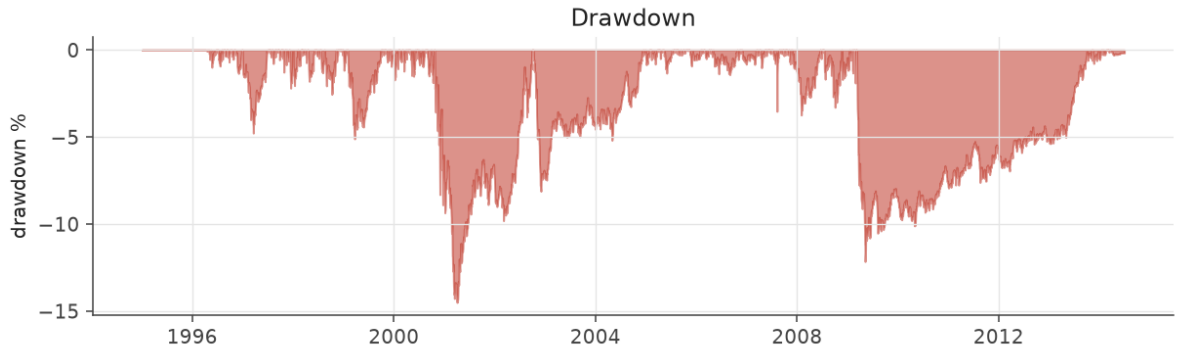
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.968

• best return among equity: `market_boost_max` (0.15); this one: 0.0384

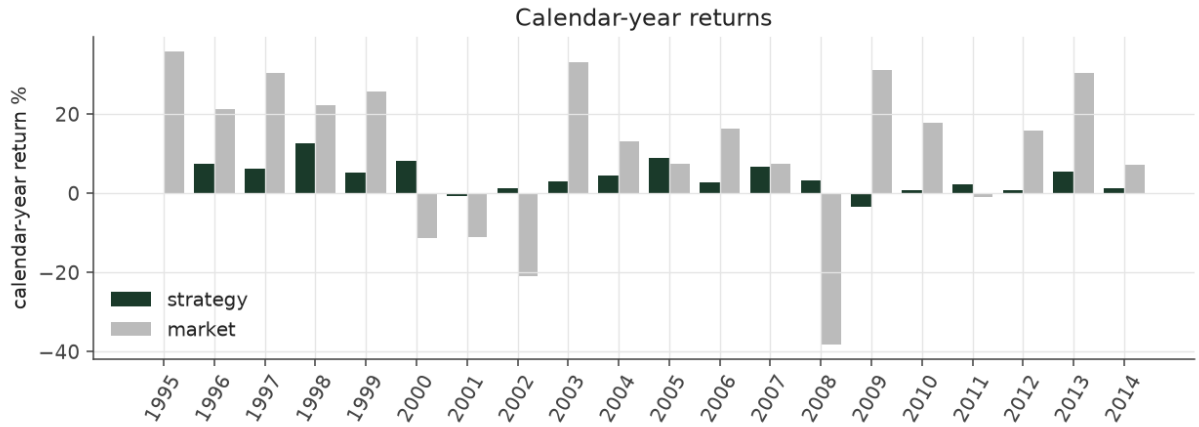
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.265



NAV (log scale)



Drawdown from running peak

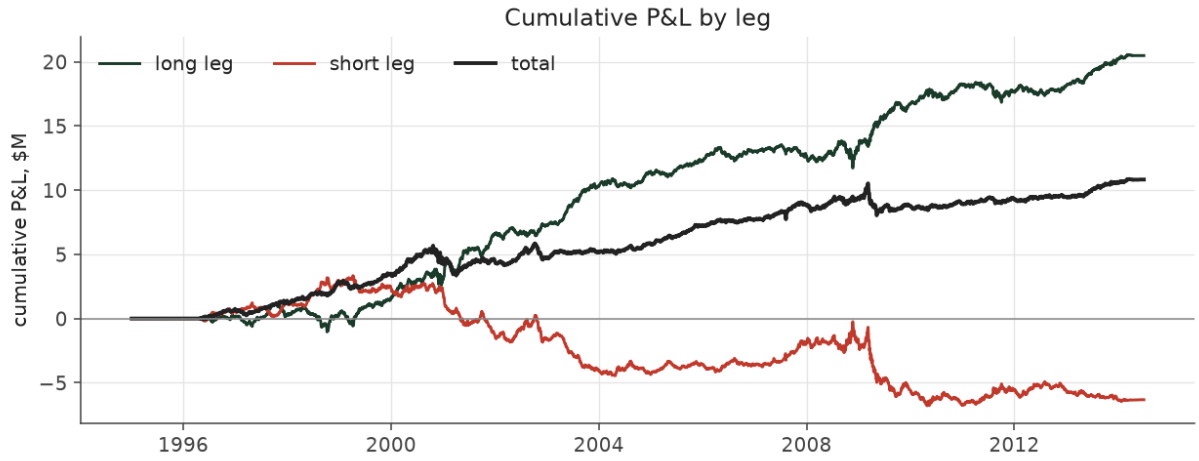


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	20.48	-6.32
P&L after costs ($M)	18.91	-8.08
of which common (average-event) component ($M)	11.70	-12.47
of which excess / PEAD component ($M)	8.78	6.15
excess minus costs ($M)	7.21	4.39
gross return per $ traded (bp)	293.19	-80.20
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.13	-0.31

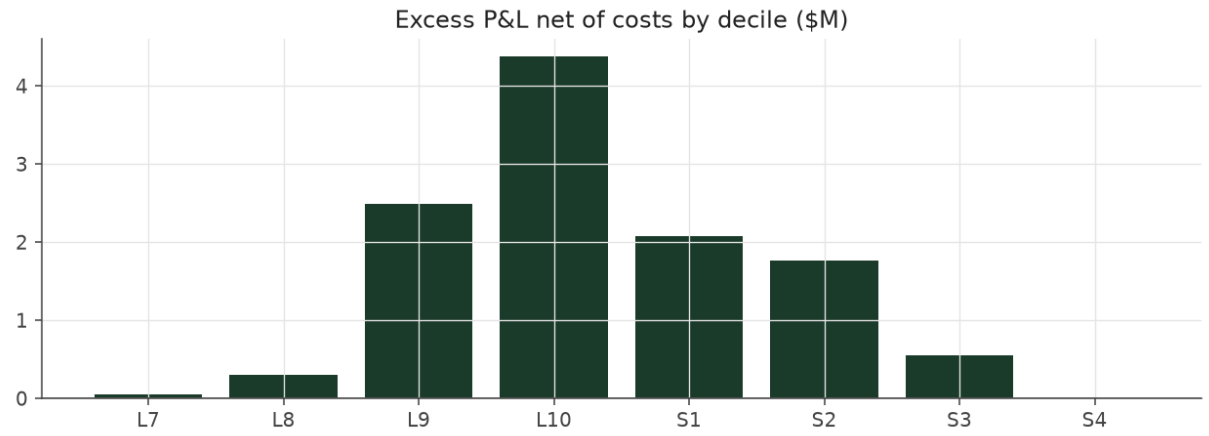
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-6.3M before costs) while the long leg earns +20.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 11.7M / 12.5M. Stripping that common component, the surprise signal itself earned +8.8M on the long side and +6.1M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.13 long / -0.31 short vs 0.97 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	1,924	18.30	0.35	0.09	0.05	0.04	0.4%	23.37
long	8	8,780	109.29	2.59	0.57	0.27	0.30	2.6%	27.46
long	9	14,436	219.84	5.74	2.98	0.50	2.49	21.4%	113.20
long	10	18,880	351.14	10.23	5.13	0.76	4.38	37.7%	124.64
short	1	17,679	366.43	-3.92	2.92	0.84	2.08	17.9%	56.74
short	2	17,172	285.54	-2.73	2.39	0.62	1.77	15.2%	61.89
short	3	9,826	126.90	-1.32	0.82	0.27	0.54	4.7%	42.86
short	4	989	9.50	-0.11	0.02	0.02	0.00	0.0%	4.24



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	11,157	231.59	1.49	2.63	0.90	1.74	15.0%	74.99
2	18,859	329.46	4.65	5.53	0.95	4.58	39.5%	139.07
3	20,115	322.68	2.79	3.56	0.68	2.88	24.9%	89.39
4	20,294	308.44	1.44	2.12	0.48	1.64	14.2%	53.26
5	19,261	294.77	0.46	1.08	0.33	0.76	6.5%	25.66

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	9,902	164.57	2.57	3.55	0.82	2.72	23.5%	165.47
2	19,150	356.76	3.04	4.53	1.07	3.46	29.8%	96.92
3	20,400	342.39	2.99	3.88	0.69	3.20	27.6%	93.36
4	20,497	322.78	1.48	2.04	0.45	1.59	13.7%	49.28

5	19,733	300.32	0.75	0.93	0.30	0.63	5.4%	20.89

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	28,529	456.20	0.67	4.90	0.96	3.93	33.9%	86.15
41-60d	22,791	384.98	0.37	2.19	0.90	1.29	11.1%	33.48
61-90d	30,360	502.34	5.04	3.68	1.04	2.64	22.7%	52.50
>90d	8,006	143.42	4.75	4.17	0.42	3.75	32.3%	261.16

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	26,048	385.15	4.33	5.49	0.82	4.66	40.2%	121.03
2002-07	30,088	512.64	4.20	5.51	1.13	4.38	37.7%	85.41
2008-13	33,550	589.14	2.31	3.94	1.38	2.56	22.1%	43.49

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	16,984	287.05	2.15	1.63	0.65	0.98	8.5%	34.26
Chems	1,801	28.30	0.08	0.22	0.06	0.16	1.4%	56.52
Durbl	2,142	35.56	0.33	0.54	0.08	0.45	3.9%	126.95
Enrgy	4,375	71.56	0.70	1.03	0.15	0.88	7.6%	122.82
Hlth	8,573	146.42	1.01	0.99	0.37	0.62	5.4%	42.40
Manuf	9,128	147.16	0.74	1.47	0.32	1.15	9.9%	77.97
Money	12,717	211.84	1.21	2.65	0.46	2.19	18.9%	103.39
NoDur	3,879	61.42	0.39	0.48	0.13	0.35	3.0%	56.66
Other	13,824	231.05	2.13	2.90	0.56	2.34	20.2%	101.42
Shops	9,776	158.08	2.03	2.35	0.35	2.00	17.2%	126.42
Telcm	3,058	52.34	0.21	0.51	0.11	0.40	3.4%	76.14
Utils	3,429	56.15	-0.15	0.18	0.10	0.08	0.7%	14.54

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	30,116	494.54	1.34	2.64	0.99	1.65	14.2%	33.29
low vol	29,650	492.29	4.60	6.32	1.22	5.10	43.9%	103.55
mid vol	28,363	468.73	4.25	5.34	1.05	4.30	37.0%	91.65

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	3.84%	0.00%
0.5%	0.8	3.63%	0.21%
1.0%	1.6	3.41%	0.43%
2.0%	3.3	2.94%	0.90%
5.0%	8.1	1.23%	2.61%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.